

Ratio & Proportion

Test · 26 marks total

Name: _____

Date: _____

Score: _____ / 26

Section A — Simplifying ratios

A1. Simplify the ratio 16:20.

[1 mark]

A2. Simplify the ratio 45:18.

[1 mark]

A3. Write the ratio 5:12 in the form 1:n, giving n to 2 decimal places.

[2 marks]

A4. A bag contains red and blue counters in the ratio 3:7. What fraction of the counters are blue?

[1 mark]

Section B — Sharing in a given ratio

B1. Share £84 in the ratio 2:5.

[2 marks]

B2. Share 120 sweets in the ratio 1:2:3.

[3 marks]

B3. Two amounts are shared in the ratio 3:4. The larger share is £48. Find the total amount shared.

[2 marks]

Section C — Direct proportion

C1. 5 pens cost £3.50. Find the cost of 8 pens.

[2 marks]

C2. A recipe for 4 people needs 200 g of rice. How much rice is needed for 10 people?

[2 marks]

C3. Shop A sells 6 apples for £2.40. Shop B sells 10 apples for £3.80. Which shop offers better value? Show your working.

[3 marks]

Section D — Inverse proportion

D1. It takes 4 workers 15 hours to build a wall, all working at the same rate. How long would it take 6 workers?

[3 marks]

D2. y is inversely proportional to x. When $x = 5$, $y = 12$. Find y when $x = 3$.

[2 marks]

D3. A car travels a fixed distance. At 40 mph the journey takes 3 hours. How long would it take at 60 mph?

[2 marks]